

DETAILED DESCRIPTION OF THE INVENTION — PRIVACY PRESERVATION, DATA OWNERSHIP, AND PERMISSION MANAGEMENT SYSTEMS

The intraluminal physiologic monitoring device and associated computational systems may incorporate privacy preservation and data governance frameworks designed to ensure secure handling of sensitive physiologic information. Such systems may enable users to maintain ownership and control over collected data while supporting meaningful analysis and ecosystem participation.

In various embodiments, physiologic data may be transformed into abstracted representations prior to storage or transmission. Raw sensor measurements may remain localized to the device or be encrypted before leaving trusted hardware environments. Data minimization strategies may limit retention of information unnecessary for analytic purposes.

Ownership models may designate users as primary custodians of physiologic data. Systems may allow users to grant, modify, or revoke access permissions governing data sharing with clinicians, research institutions, partners, or authorized third-party services.

Permission management architectures may employ token-based authorization, encrypted identity credentials, biometric verification, or decentralized identity frameworks enabling secure authentication without exposing personal information.

Privacy-preserving computation methods may include federated learning, secure multiparty computation, homomorphic encryption, differential privacy techniques, or anonymized aggregation allowing population-level insights to be generated without disclosure of identifiable user data.

Data access controls may define hierarchical permission levels including personal viewing, partner sharing, clinical supervision, research participation, or automated system optimization. Permission status may be adjustable dynamically through user interfaces.

Audit logging systems may record data access events, authorization changes, and system interactions to provide transparency and accountability. Users or administrators may review access history to verify compliance with privacy policies.

In certain embodiments, users may export, transfer, archive, or permanently delete physiologic datasets in accordance with data portability principles and jurisdictional privacy regulations.

The privacy preservation and data ownership mechanisms described herein are illustrative rather than limiting. Any architecture enabling secure data control, user ownership, and permission-managed sharing of intraluminal physiologic monitoring data is considered within the scope of the present disclosure.